

EE-305 Electromagnetic Theory (EE-1, 2)

Mid Term – 1(Fall 2016)

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Dated: September 21, 2016

INSTRUCTIONS: Do not ask any question from the invigilator, make a valid assumption in case of any ambiguity. Attempt all questions with proper numbering on the answer sheet. Your work should be neat and clean.

Total number of pages: 2

Question # 1 [8+12= 20 marks]

(a) Define scalar and vector fields with two real physical examples each.

(b) A vector field $\vec{T}(\vec{r})$ is expressed in rectangular coordinates as:

$$\vec{T}(\vec{r}) = \left\{ \frac{125}{[(x-1)^2 + (y-2)^2 + (z+1)^2]} \right\} \{ (x-1)\vec{a}_x + (y-2)\vec{a}_y + (z+1)\vec{a}_z \}$$

(i) Evaluate the field $\vec{T}$ at $P(2, 4, 3)$.

(ii) Determine a unit vector that gives the direction of $\vec{T}$ at P .

(iii) Specify the surface $f(x, y, z)$ on which $|\vec{T}| = 1$.

Question # 2 [15+15=30 marks]

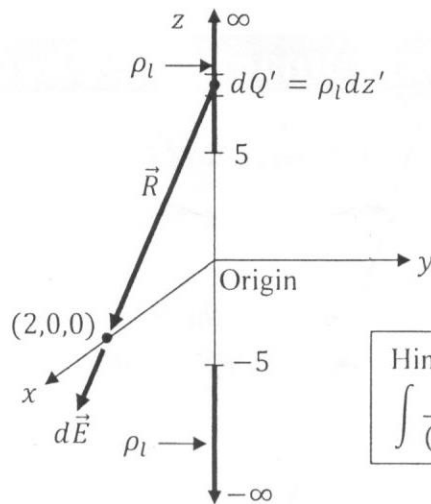
(a) Short Questions:

(i) State Coulomb's law for two point charges and its vector representation with the help of diagram and mathematical expression.

(ii) Also write the expressions of force exerted on a charge 'q' by 'N' point charges and the force exerted by volume charge distribution ' ρ_v ' [C/m^3].

(iii) A volume charge density of $\rho_v = e^{-2r}/r^2$ is located in universe. Identifying the coordinate system, calculate the total charge within the universe

- (b) A uniform line charge is distributed along the z -axis from $z = 5\text{m}$ to ∞ and from $z = -5\text{m}$ to $-\infty$ with a density of $\rho_l = 20\text{ nC/m}$. Find the force $\vec{F}$ exerted on a test charge 'q (infinitesimally small)' at $(2, 0, 0)\text{m}$ in Cartesian (rectangular) coordinates.



Hint:

$$\int \frac{dx}{(a^2 + x^2)^{3/2}} = \frac{x}{a^2 \sqrt{a^2 + x^2}}$$